Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

MAKEUP EXAMINATIONS, DECEMBER 2024

SWITCHGEAR AND PROTECTION [ELE 3123]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 01 JANUARY 2025** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A	Compare the axial, cross-jet, and radial blast types of air-blast circuit breakers in terms of construction, working, and application	***(04)***
1B	Explain the operating mechanisms of circuit breakers and describe the maintenance procedure for GIS (Gas Insulated Switchgear).	***(03)***
1C	In a system of 132 kV, the circuit phase to ground capacitance is 0.01 μF, and the inductance is 6 Henrys. Calculate the voltage across a circuit breaker's pole if a magnetizing current of 5 amps (instantaneous value) is interrupted. Calculate the value of resistance to be used across the contacts to eliminate the restriking voltage.	***(03)***
2A	How does capacitive-current interruption influence transient overvoltages in power systems, and what mitigation strategies can be employed?	***(03)***
2B	Explain the role of isolators and earthing switches in the safe isolation of a circuit and their sequence in a switching operation with circuit breakers.	***(03)***
2C	Prove that the fault current (IF) is 3 times the normal current in the ungrounded systems with the help of the relevant circuit and phasor diagram. Also, Mention the disadvantages of ungrounded systems in power systems.	***(04)***
3A	Discuss the Essential qualities required for a protective relay	***(04)***
3B	With the help of the one-line diagram of the primary relaying scheme of a typical power system, explain the Zones of Protection.	***(03)***
3C	Derive an expression for torque produced by an electromagnetic induction type relay.	***(03)***
4A	With the help of the connection diagram, explain the working principle of Merz- Price protection for Star-Delta Transformer.	***(03)***
4B	Describe the consequences of failure of the prime mover to an alternator and a suitable protection scheme. Justify the statement: "Protection provided to the alternator against failure of prime mover is for the prime mover rather than the alternator".	***(03)***

4C. Determine three zone relay settings (referred to the secondary of CT and PT) for the distance protection of transmission line assuming

i. All 3 zones using impedance relay units

ii. All 3 zones using reactance relay units

iii. All 3 zones using mho relay units with a characteristic angle of 50°

Use the following data to determine relay settings.

- CT ratio: 400/1 A
- PT ratio: 132kV /110V
- Impedance of line 1: (2.5+ j5.5) Ω
- Impedance of line 2: (3 + j6.2) Ω
- Impedance of line 3: (3.5 + j7) Ω
- Zone 1 covers 80% of line 1
- Zone 2 covers line 1 and 40 % of line 2
- Zone 3 covers line 1, line 2 and 15% of line 3. ***(04)***

5A. With the help of relevant circuit schematic, describe the Translay scheme of wire pilot protection to an overhead transmission line. Discuss the limitations of this scheme. ***(04)***

5B. What are the consequences of overloading a 3-phase Induction motor? Describe the thermal overload protection using a replica relay. ***(03)***

5C. Discuss in detail the advantages of numerical relaying over electromechanical and static relays. ***(03)***